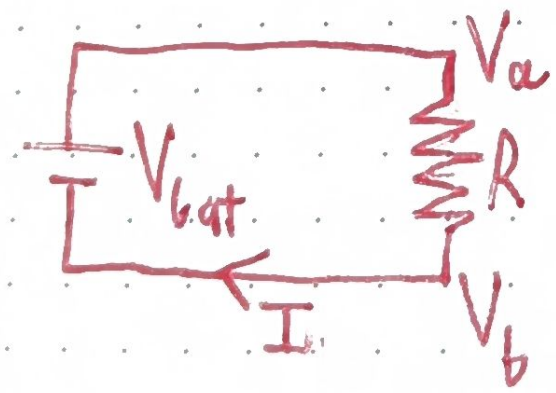
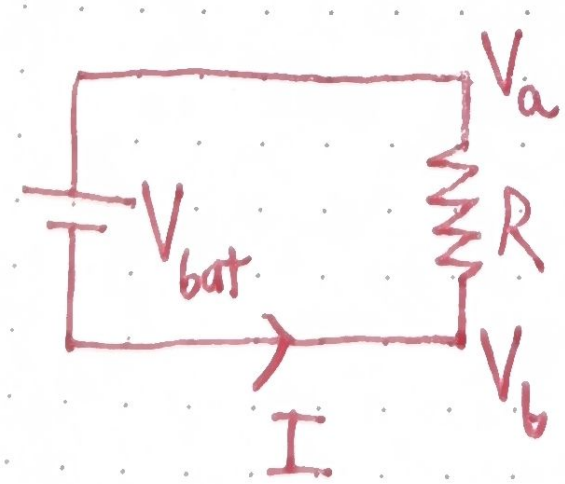
250908 Ohm's Law

- Ordinary Node: A Point that Connects <u>ONLY TWO</u> elements
- Extraordinary Node: A node connected to 3 or More elements

- Loop: A Closed Path where the Start and end nodes are the Same.
- Mesh: A loop that encloses no other loop.



$$I = \frac{V_a - V_b}{R} = V_{bat}/R$$



Current Start

Current End

$$I = \frac{V_b - V_a}{R} = -V_{bat}/R$$

Power:

From $V = IR$

$$P = IV = I^2 R = V^2/R$$

$$[J/s] = \left[\frac{C}{s}\right]\left[\frac{J}{C}\right] \checkmark$$

From $I = \frac{V}{R}$

Power prop-to I or V squared

Energy:

$$W = P\Delta t \qquad [J] = [J/s][s] \checkmark$$

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ex/ Take a $\boxed{10\ A\cdot hr}$, 10 V Battery w/ a 1 W load

Unit of Charge

- Find Current:

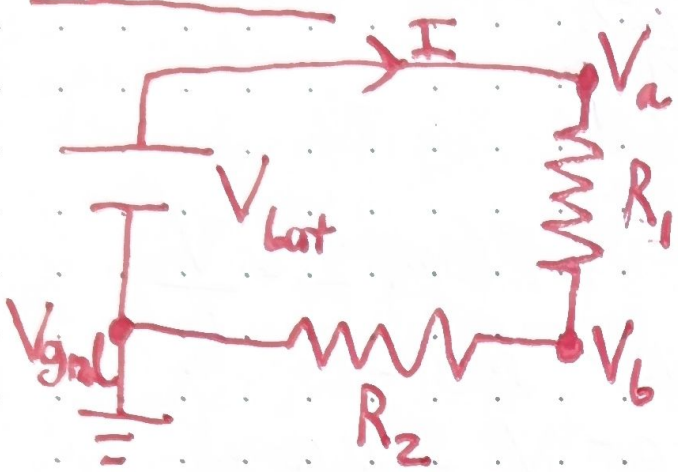
$$P = IV \Rightarrow 1W = I(10V)$$
$$\Rightarrow I = 0.1\ A$$

- Dimentional Analysis:

$$(10\ \cancel{A}\cdot hr)\left(\frac{1}{0.1\cancel{A}}\right) = 100\ hr$$

$\therefore$ The battery will last 100 hr

Series:



set $V_{gnd} = 0 \Rightarrow V_a = V_{bat}$

Look at R_1:

$$I = \frac{V_a - V_b}{R_1} = \frac{V_{bat} - V_b}{R_1}$$

Look at R_2: $I = \dfrac{V_b - V_{gnd}}{R_2} = V_b / R_2$

$\Rightarrow V_b = I R_2$

Thus: $I = \dfrac{V_{bat} - I R_2}{R_1} \Rightarrow I R_1 = V_{bat} - I R_2$

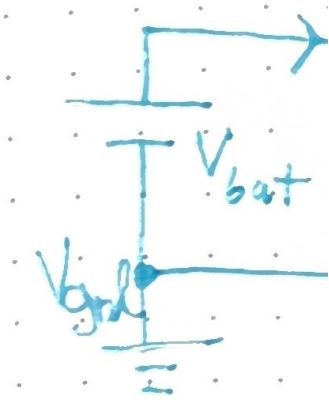
$\Rightarrow I R_1 + I R_2 = V_{bat} \Rightarrow I(R_1 + R_2) = V_{bat}$

let $R_{eq} = R_1 + R_2$

$\Rightarrow V_{bat} = I R_{eq}$

In general $\boxed{R_{eq,series} = \sum_i R_i}$

Parallel



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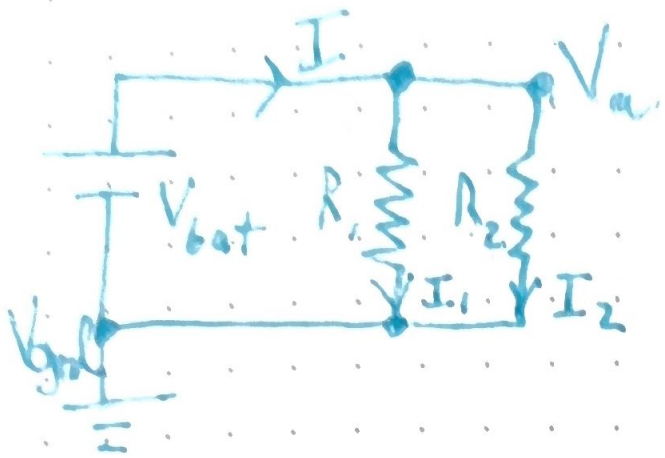
Parallel: Set $V_{gnd} = 0$

$\Rightarrow V_a = V_{bat}$



Look at R_1:

$$I_1 = \frac{V_a - V_{gnd}}{R_1} = V_a / R_1 = V_{bat} / R_1$$

Look at R_2: $I_2 = \dfrac{V_a - V_{gnd}}{R_2} = V_a / R_2 = V_{bat} / R_2$

Look at node a: $I = I_1 + I_2$

$$\Rightarrow I = \frac{V_{bat}}{R_1} + \frac{V_{bat}}{R_2} \Rightarrow I = V_{bat}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$\Rightarrow V_{bat} = I\left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1}$$

let $R_{eq} = \left(\dfrac{1}{R_1} + \dfrac{1}{R_2}\right)^{-1}$

$$\Rightarrow V_{bat} = I R_{eq}$$

In general $\boxed{R_{eq,\,parallel} = \left(\sum_i \frac{1}{R_i}\right)^{-1}}$